

# Capital Bikeshare Specification Report

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**Abstract** – This report explores the analysis of Capital Bikeshare service, a major bike sharing firm in Washington, to evaluate a critical goal to increase subscription enrolment for new customers, feasibility to expand dock stations in response to rising demand for sustainable transportation alternatives for environmental and health benefits. The projected expansion of dock stations intends to solve current restrictions while capitalizing on new trends in urban transportation, with an emphasis on improving accessibility, user experience, and system efficiency. An in-depth evaluation of crucial elements such as user behavior, geographic demand patterns, and the dynamic landscape of city planning is included in the analysis. The research also discusses the possible benefits of increasing dock stations, such as increased service coverage, improved customer convenience, and increased system resilience. The advantages of increasing the number of dock stations for Capital Bikeshare emerge as a critical strategy for positioning the company as a leader in sustainable urban mobility. The findings in this report are intended to help decision-makers understand the implications, benefits, and potential obstacles connected with this strategic undertaking.

**Keywords** – *Bike sharing, sustainable, Intelligent transport system, SDG11*

## I. INTRODUCTION

### A. Organisation Background & Marketplace

Capital Bikeshare is a prominent bicycle sharing company officially launched in September 20, 2010 with the initial rollout in Washington, D.C. The program was a joint effort of the District of Columbia, Arlington County, the City of Alexandria, Montgomery County, and Fairfax County. Capital Bikeshare is very much focused in promoting sustainable transportation alternatives, reducing traffic congestion, and reducing the environmental impact of traditional commuting methods. The program contributed to the region's goals of fostering a greener and more eco-friendly urban environment.

This bikeshare service is in accordance with the Sustainable Development Goals 11- Sustainable cities and Communities, where cycling makes cities and human settlement be inclusive, reliable, safe and sustainable as cycling is affordable, non-polluting and promotes a sustainable economy.

The first phase of deployment included strategically placed bike stations in key areas throughout the participating jurisdictions. These stations were outfitted with short-term rental bicycles, providing users with a last-mile transportation solution which is available round the clock.

Capital Bikeshare expanded its presence in subsequent years by adding more stations and increasing the number of available bicycles. The program's success prompted increased collaboration with neighbouring jurisdictions, resulting in a regional bike-sharing network. Capital Bikeshare's user adoption increased significantly, attracting a diverse range of riders, including commuters, tourists, and recreational users.

The program's popularity was boosted by the convenience of on-demand bicycles and the ease of access.

Capital Bikeshare has incorporated technological advancements to improve user experience over the years. With the introduction of a mobile app, users could more easily locate available bikes, check station statuses, affordable membership options and plan their routes. Now the company wants to increase number of docks in existing stations in Arlington area to attract more customers, construct new stations in strategically important connecting areas and to enable access to charging stations for privately owned e-bikes. With this the users who owns personal bikes also can use docking stations to charge their bikes during their commute.

### B. Marketplace Participation

Capital Bikeshare operates in the urban transportation and sustainable mobility solutions markets, with a particular emphasis on short-distance travel within the Washington, D.C. metropolitan area. The program addresses the changing needs of city dwellers, commuters, and visitors looking for convenient, environmentally friendly alternatives to traditional modes of transportation.

Capital Bikeshare competes with the entities like traditional Bicycle rentals, public transportation which includes buses and trains. Also, with already existing ride sharing services like Uber, Lime and Lyft. Capital Bikeshare has an advantage over the traditional modes due to its 24/7 availability, ease of use and convenience makes the user prefer bikeshare as an ideal alternative.

## II. SCOPE OF PROCESS

Capital Bikeshare must consider and optimize several processes in order to operate effectively in the above-mentioned marketplace:

### A. Station planning and placement

All the stations should be planned strategically and to be placed at hotspots of the cities by considering the factors where peoples density is high, shopping zones, public gatherings and people's commuting patterns.

### B. Bicycle Distribution and Maintenance

To meet demand, processes for the efficient distribution of bicycles across stations are being implemented. Creating detailed maintenance procedures to ensure the fleet's reliability and longevity.

### C. User Signup and Access

Streamlining the user registration process, making it simple for people to sign up and get bikes quickly. Membership plans should be attractively designed for users to pick from a weekly, monthly, quarterly, semi-annual, or annual pass. In addition, membership renewals should be paid via auto debit. Some users may forget to renew their membership on time and may renew it when they use the service next time. Due to this, the user will leave the customer

lifecycle and necessary efforts need to be brought to hold back the users in the membership plan. However, enabling the auto renewal facility will assist the company in receiving the membership fee on time, thereby reducing revenue leakage and provide latest updates and benefits from company's newsletters.

#### D. Technology Integration

Capital Bikeshare enhances user experiences by utilizing cutting-edge technologies such as GPS, IoT devices, and mobile apps. These advancements allow for real-time tracking of bike availability, ensuring users have easy access. Furthermore, the incorporation of advanced technologies improves operational efficiency, enabling dynamic bike distribution and streamlined maintenance. Capital Bikeshare's commitment to technological advancement underscores the company's commitment to providing a seamless, user-centric, and sustainable urban transportation solution.

#### E. Scalability Planning

Capital Bikeshare scalability planning entails anticipating and accommodating the program's growth. The identification of new station locations to meet changing demand is guided by an assessment of potential increases in user counts. Adapting to changing urban dynamics ensures that the system is in sync with city growth. This strategic approach considers not only increased usage but also the integration of emerging technologies and innovations. Capital Bikeshare demonstrates its commitment to providing a robust, responsive, and sustainable bike-sharing solution for expanding urban communities by envisioning scalability. Furthermore, each dock station should have more optional bike stands to accommodate a greater number of bikes during peak hour if more users dropped the bikes in a particular station.

#### F. Community Partnership

Capital Bikeshare actively seeks collaborations with local businesses, institutions, and events to enrich its presence and accessibility. Partnerships with businesses may involve co-branded promotions, offering employees convenient commuting options. Also, erecting stations inside the parking lot of shopping zone not only helps the local business to sell their products, it is also convenient for users to carry the products in the bike's basket. Aligning with institutions can lead to tailored programs for students or faculty. It also helps the workers whose work hours end during the night time when no one transportations facility is available. They can use this service without waiting for anything and reach their home easily and quickly compared to other services. In event collaborations, special promotions during festivals or gatherings enhance user engagement. These partnerships not only enhance the program's visibility but also create a symbiotic relationship, contributing to the community's vibrancy while offering diverse user groups convenient and sustainable transportation options.

### III. BUSINESS VALUE

#### A. Sustainable Transportation

Capital Bikeshare promotes eco-friendly and sustainable transportation, reducing carbon emissions and contributing to a greener environment.

#### B. Urban Mobility

Provides a convenient and efficient solution for urban mobility, reducing traffic congestion and offering an alternative mode of transportation.

#### C. Last Mile Connectivity

Serves as a last-mile connectivity option, complementing public transportation systems and making it easier for users to reach their final destinations.

#### D. Health and Wellness

Encourages physical activity and a healthier lifestyle by providing an accessible and active means of commuting.

#### E. Corporate Social Responsibility

Demonstrates a commitment to CSR by promoting sustainable and responsible business practices that benefit both the community and the environment.

### IV. PORTER FIVE FORCES

Michael Porter created the Porter's Five Forces framework to analyse the competitive forces within an industry. While the original model was created for traditional industries, it can be applied to a variety of industries, including the bike share industry. Here's an example of how Porter's Five Forces can be applied to a bike share company:

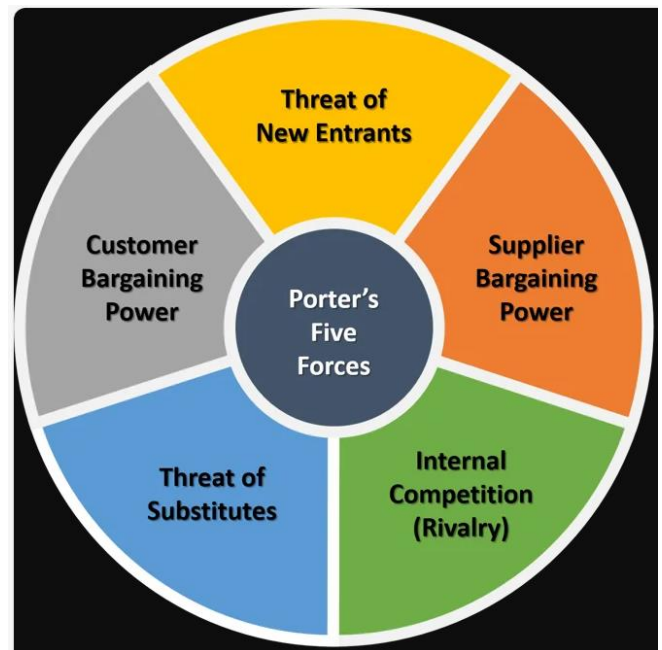


Fig. 1. Porter Five Forces

#### A. Threat of new entrants

- **Barrier to entry:** High initial investment is needed as upfront for research, development, manufacturing, marketing and establishing proper distribution network. Existing companies may benefit from economic of sales, which is difficult for new entrant to match production and cost effectiveness. Since the new entrants has limited resources, they won't able to compete with the existing competitors. Also, new entrants may face difficulties in securing relationship with established retailers, limiting their access to consumers.

- **Government Regulations:** Government regulations play a pivotal role in shaping the electric bike sharing industry, with direct implications for both established players and new entrants. Safety standards imposed by the regulatory bodies ensure customer protection and product reliability. Compliance with these standards is not only required by law, but it is also necessary for establishing consumer trust. New entrants may face difficulties in maintaining the complex standards of addressing issues like energy efficiency and recyclability, which have an impact on manufacturing processes & product design, guiding the industry toward sustainability.

These factors make the new entrants not a threat to the existing organisations.

#### *B. Supplier bargaining power*

- **Supplier Concentration:** It is a critical factor influencing the dynamics of bike sharing industry. When there is only limited supplier who manufactures the essential components, they become the king and rule the industry with lots of influence over manufactures. This will lead to higher prices, reduced flexibility and disruption over the supply chain. Diversified suppliers and strategic partnership helps the manufactures to lower the supplier concentration and maintain competitive edge in the field.
- **Unique Resources:** Supplier's bargaining power increases significantly when they provide resources that are unique, irreplaceable, or possess specialized expertise. If there are limited options for the crucial components, then supplier has more power over the manufactures. To overcome this manufacturer must carefully assess the uniqueness of resources provided by suppliers and devise strategies to diversify sources or negotiate favourable terms, ensuring resilience and mitigating potential vulnerabilities associated with supplier concentration of power.

#### *C. Internal Competition (Rivalry)*

- The evaluation of the number and strength of competitors is a key aspect in understanding the competitive landscape of the bike share market. A high number of competitors intensify the rivalry. The strength of the competitors is measured by the brand value, market share, product innovation and effective distribution system. The intensity of competition makes significantly affect the market dynamics and profit. In a crowded field, to attract the customer's attention the companies should drive them innovatively and make them unique compared to others.
- **Industry Growth:** The overall growth of the organisation should be measured. In case the growth the organisation is slow, the demand will be very low. The companies will fight for their share to achieve this they will have aggressive marketing strategies, competitive prices and high innovation to acquire their shares. Manufactures should differentiate their products, enhance the efficiency, and stand out of the crowd to make them unique.

#### *D. Threat of Substitutes*

- **Substitutability:** The substitutes for bike sharing system is Traditional bicycles, whether owned or rented, are a traditional substitute for short-distance travel. Electric and manual scooters are another convenient option. Emerging personal mobility devices, such as electric skateboards or hoverboards, also offer alternatives. The accessibility and appeal of these alternatives have a significant impact on the competitiveness of bike-sharing services. Capital Bikeshare must enhance continually its service offering, user experience to keep it as a preferred choice.
- **Technological Advances:** Continuous innovation in the field of bike sharing technology will be led to new and more attractive alternatives, which will make the current transportation less effective. Evolution of AI has introduced the autonomous vehicles and smart city initiatives can significantly influence customers preferences. Advanced technologies should be implemented to increase the user experience by offering convenient alternatives. This will make the Capital Bikeshare to remain adaptive and always available on top of the market.

#### *E. Customer Bargaining power*

- **Customer Loyalty:** Strong brand loyalty, in which users repeatedly choose and prefer the bike sharing service, can reduce buyers bargaining power. A loyal customer base is less likely to be altered by new services or price increases, giving the company a sense of stability and resilience. It also increases the likelihood of customer retention and positive word of mouth, which contributes to long term growth. To cultivate and strengthen loyalty and maintain a strong position in the competitive landscape, Capital Bikeshare must invest in customer satisfaction initiatives, personalized experiences, and continuous service improvements.
- **Price Sensitivity:** Evaluating price sensitivity needs deep understanding of customer behavior and its impact on bargaining power. If a customer is sensitive to price, they will more likely to look for alternate options, switch service provider and negotiate for better deals in response of price change. So, Capital Bikeshare should take this into considerations before increasing the prices, make sure it should not affect the price sensitive customer base. To maintain a favorable position and customer satisfaction in price-sensitive environments, businesses must be aware of market dynamics, use effective pricing models, and consider value-added offerings.

### **V. BALANCED SCORE CARD**

#### *A. Customer Aspects*

- **Initiative:** Conduct frequent customer satisfaction surveys to perceive customer characteristics, benefits of using bike sharing system and

knowledge on road safety measures to travel without any accidents.

*Measure:* It is measured in aspects of customer satisfaction.

*Target:* To achieve happier customers continually using Capital Bikeshare service.

- *Initiative:* Enable services for customers to choose Capital Bikeshare as an ideal choice for shorter commute.

*Measure:* This is measured in the aspects of purchase frequencies.

*Target:* To achieve repeated customers.

- *Initiative:* Bring customers under Capital Bikeshare platform by providing membership plans targeting all customer segments.

*Measure:* It is measured in the aspects of average revenue per user.

*Target:* To become a leading Bikeshare company in Washington region.



Fig. 2. Balanced Score card

### B. Financial Aspects

- *Initiative:* Strategically placing membership plans to find constant revenue from loyal customers.

*Measure:* It is measured in product revenue.

*Target:* To achieve positive year over year growth.

- *Initiative:* Bring down maintenance cost for cycle and stations, Operational costs and overheads.

*Measure:* Measured by Net Profit Margin.

*Target:* Achieve 20% reduction in overall cost for maintenance and operations.

- *Initiative:* Marketing and Expansion: Allocate funds to promote the presence of Bikeshare and increase the fleet.

*Measure:* by Customer Acquisition Cost.

*Target:* To increase the membership.

### C. Internal Process

- *Initiative:* Enhance customer service experience through efficient process.

*Measure:* Measured by the response time to customer queries, customer issues resolution rate and customer satisfaction rate.

*Target:* To achieve first class services and prompt response to address customer queries and grievances. Increased customer satisfaction.

- *Initiative:* Ensure the high quality of the output and service is delivered.

*Measure:* Measured by defect rates, customer complaints resolution time and adherence to quality aspects.

*Target:* To achieve reduced defect rates with high quality products, which lead to less customer complaints and reduced resolution time.

### D. Learning and Growth

- *Initiative:* Foster continuous improvement and innovation. The ability to adapt, learn, and grow is crucial for long-term success.

*Measure:* Measured based on number of trainings provided to the employee.

*Target:* Employees are trained to the new technology. So, they can work independently without causing issues.

- *Initiative:* To develop positive work environment and job satisfaction.

*Measure:* Measured by employee satisfaction survey, employee retention ration, attendance and punctuality of the employee.

*Target:* To achieve satisfied employee, working with 100% effort.

## VI. SYSTEM DESIGN

### A. Process flow diagram

The below process flow diagram explains that the user logs in the system if the user exist in the system, it will be redirected to the Bike booking page or else the app redirected to the sign-up page. Once the user logs in, the user can book bike from the list of bikes available. Once the bike is booked, the user has option to cancel the bike if the user changes the commute plan. If the user wants to continue the commute, he can start the ride from the start station that was mentioned during bike and can reach the destination station of their choice. Once the user destination station is reached, the user can park the bike and pay the ride charges as per the duration. All these details are stored in the relational database.

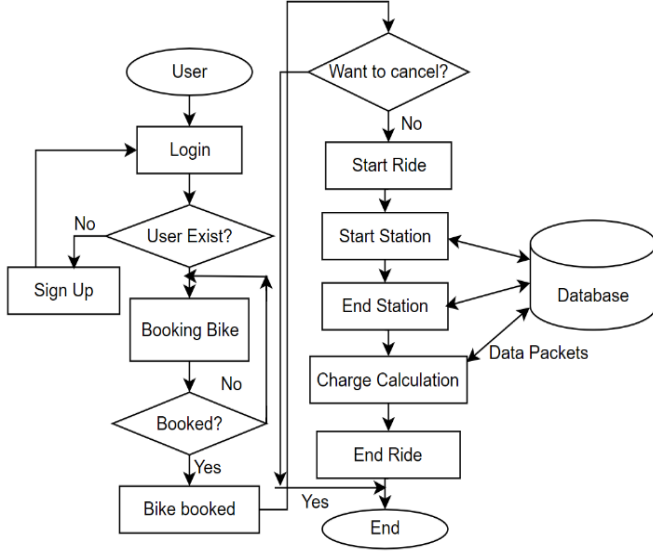


Fig. 3. Process flow diagram

### B. Data Capture Points

The data collection points provide critical information for a variety of purposes, such as user activity analysis, ride statistics, billing and invoicing, and system performance evaluation. Furthermore, location data, timestamps, and user IDs are extremely useful for improving user experience and optimizing system operations.

TABLE I. DATA CAPTURE POINTS

| Data Capture points |                                                            |
|---------------------|------------------------------------------------------------|
| Capture point       | Data                                                       |
| Login               | Sign in information of the user with membership details    |
| Start Ride          | Capture start station details and geographical coordinates |
| End Ride            | Capture end station details and geographical coordinates   |
| Charge Calculation  | Calculating the duration of ride and bill amount           |

### C. Analytical Requirements

The process of studying & comprehending the customer and their requirements in order to define the problem domain and system requirements.

- **User Behaviour analysis:** The objective of user behaviour analysis is to understand patterns and preferences of bike usage. Visualization will be done for ride frequency, average ride duration, peak usage times, rider types based on gender.
- **Station utilization Metrics:** The objective of the station utilization metrics is to optimize popular stations, availability of cycles and docks in all the stations.
- **Bike utilization metrics:** The objective of this dashboard is to get a comprehensive understanding of the bikes in the fleet and their usage patterns.
- **Financial Performance Analysis:** The objective of the financial performance analysis is to understand revenue generated by the different bike types by the customer's age.
- **Maintenance Predictive Analysis:** The objective of the maintenance predictive analysis is to understand the current condition of the bikes and proposed dates for maintenance.
- **Competitor Analysis:** The objective of the competitor analysis is to understand the market share of competitors operating in Washington.
- **Market Expansion Analysis:** The objective of the market expansion analysis is to find the new location to construct the dock stations near the population density area or people gathering area.

### D. Customer Integration

- **User friendly mobile app:** A customer friendly mobile app has been already available, it allows the user to locate bikes, check station status and provide feedbacks.
- **In App notification:** Proactively communicate system updates, promotions, and relevant information to users.

## VII. DATABASE DESIGN

### A. Entity Relationship Diagram

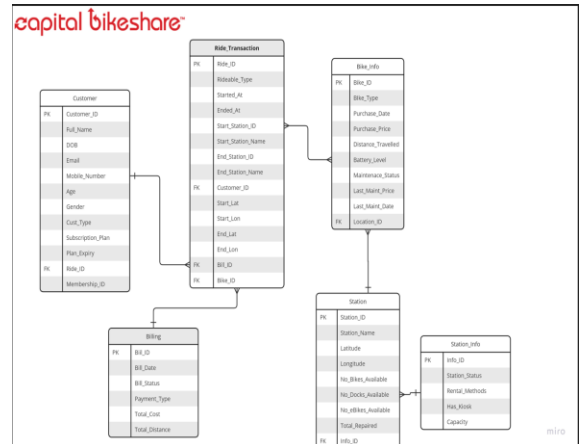


Fig. 4. Entity relationship diagram



The Entity Relationship diagram of Capital Bikeshare has 6 tables. The customer tables have 12 attributes containing the user details with 1 primary key customer ID, 1 foreign key Ride ID and the other columns contain the basic details of the customer.

The Ride transaction tables have 15 attributes, this table has one to many relationships with other tables like customer details, Bike info, billing and stations details via a foreign key. Also, this table has details of the rider information, start and end station details with their geographic coordinates and bike details.

The billing detail tables have Bill\_ID attribute as primary key which has relationship with ride transaction table by joining bill ID attributes. Also, this table has information like bill date, bill status, payment type, ride charges and total distance travelled in that particular trip.

The bike info table has a primary key called Bike\_id which has relationship with ride transaction table. Also, this table captures the complete details of bike.

The station details tables contain a primary key called station\_id which has relationship with bike information table. This table provides the details of dock stations and bike availability.

The station info table has a primary key called Info\_id which has relationship with station details table. This table provides information like how many docks can be used for private vehicles charging and their available capacity.

## B. Data Dictionary Views

| Ride Transaction   | Data_Type | Definition                          |
|--------------------|-----------|-------------------------------------|
| Ride_ID            | String    | Alphanumeric ID for bike trip       |
| Rideable_Type      | String    | Type of bike                        |
| Started At         | DateTime  | Start date and time                 |
| Ended At           | DateTime  | End date and time                   |
| Start Station ID   | Numeric   | Numeric ID for a station            |
| Start Station Name | String    | Name of the station                 |
| End Station ID     | Numeric   | numeric ID for a station            |
| End Station Name   | String    | Name of the station                 |
| Customer_ID        | Numeric   | Nuemic ID for each customer         |
| Start_lat          | Float     | Geo-coordinates for start latitude  |
| Start_lng          | Float     | Geo-coordinates for start longitude |
| End_lat            | Float     | Geo-coordinates for end latitude    |
| End_lng            | Float     | Geo-coordinates for end longitude   |
| Bill_ID            | Numeric   | Numeric ID for the bill generated   |

Fig. 5. Data dictionary for User Ride details

| Station              | Data_Type | Definition                              |
|----------------------|-----------|-----------------------------------------|
| Station_ID           | Numeric   | Numeric ID for a station                |
| Name                 | String    | Name of the station                     |
| Latitude             | Float     | Geo-coordinates for station latitude    |
| Longitude            | Float     | Geo-coordinates for station longitude   |
| Num_Bikes_Available  | Numeric   | Available number of bikes in a station  |
| Num_Docks_Available  | Numeric   | Available number of docks in a station  |
| Num_Ebikes_Available | Numeric   | Available number of ebikes in a station |
| Capacity             | Numeric   | Capacity of the Station                 |

Fig. 6. Data dictionary for Station Details

| Station Info   | Data_Type | Definition                              |
|----------------|-----------|-----------------------------------------|
| Object_ID      | Numeric   | Numeric ID for a record in station_info |
| Station_ID     | Numeric   | Numeric ID for a station                |
| Station Status | String    | Station Active or Inactive              |
| Has Kiosk      | Bool      | Station has Kiosk or not                |
| Rental Methods | String    | Methods of renting a bike               |

Fig. 7. Data dictionary for Dock status

| Customer          | Data_Type | Definition                 |
|-------------------|-----------|----------------------------|
| Customer_ID       | Numeric   | Numeric ID for a customer  |
| Age               | Numeric   | Age of the customer        |
| Gender            | String    | Gender of the customer     |
| member_Casual     | String    | Member or casual riders    |
| Subscription_plan | String    | Type of plan in membership |
| Plan_Expiry       | DateTime  | Expiry date of the plan    |

Fig. 8. Data dictionary for Customer Information

| Billing            | Data_Type | Definition                               |
|--------------------|-----------|------------------------------------------|
| Bill_ID            | Numeric   | Numeric ID for the bill generated        |
| Bill_Date          | DateTime  | Date when the bill is generated          |
| Bill_Status        | String    | Status of the bill                       |
| Payment_Type       | String    | Mode of payment                          |
| Distance travelled | Float     | Total distance travelled during the trip |
| Cost               | Float     | Bill amount generated                    |

Fig. 9. Data dictionary for Bill details

| Bike Info             | Data_Type | Definition                              |
|-----------------------|-----------|-----------------------------------------|
| Bike_id               | Numeric   | Numeric ID for Individual Bike          |
| Bike_type             | String    | Types of Bike                           |
| Purchase_date         | DateTime  | Date when the Bike Purchased            |
| Purchase_price        | Numeric   | Cost of the Bike Purchased              |
| Past_maintenance_cost | Numeric   | Total cost incurred for all maintenance |
| Maintenance_status    | String    | Maintenance Status                      |
| Last_maintenance_date | DateTime  | Date of last maintenance                |
| Last_maintenance_cost | Float     | Cost incurred for last maintenance      |
| Distance_travelled    | Float     | Distance travelled by the Bike          |

Fig. 10. Data dictionary for Bike information

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